

Effectiveness of Active vs. Inactive Teaching Methods Supported by AI-Generated Imagery: Fostering Sedimentary Rock Understanding in Children



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Abstract:

This study explores the impact of combining active, hands-on activities with rock samples and AI-generated, child-friendly visual aids, to enhance K-2 students' ability to identify sedimentary rocks. After receiving a brief lecture on rock types, students participated in interactive, tactile demonstrations with real rock specimens and viewed AI-generated, cartoon-style illustrations designed for their reading level. The students of each class, (27 second graders and 47 kindergartners), were then assigned a four-question quiz meant to test their identification skills of sedimentary rocks following the lesson: two questions involved images of rocks the students had physically explored with activities (**active learning**), while the other two involved images of rocks they had not encountered in a hands-on context (**inactive learning**). The results of the questionnaires showed that for the "inactively" taught examples, the students achieved accuracy rates of 45% and 59% in correctly identifying the sedimentary rock. However, the students' performance on "actively" learned examples, yielded accuracy rates of 79% and 86%. These findings suggest that incorporating engaging experiences alongside age-appropriate AI visuals can enhance young learners' ability to identify geological materials. Additionally, the observed variation in scores among students of different grades indicate a need for further research into the factors influencing experiential learning outcomes in early science education, especially in multimodal, AI-supported teaching approaches.

Methodology + Materials:

1. Our agenda included briefly lecturing the students about mountains and rocks, specifically sedimentary rocks and their multiple kinds of formation and characteristics. This part of the lesson was backed up with cartoons and AI images to better reinforce what we were explaining verbally.
2. The next part of our lesson was interactive, where we visualized our points with hands-on activities. With playdoh, beads, and paper leaves, students formed their own mini sedimentary rocks that encompassed the process of compaction and fossilization.
3. Our next activity involved the children creating mountains by pushing cardboard upwards with their feet. They also simulated earthquakes and tectonic activity by placing the cardboard on top of their heads and moving slowly around the classroom. They also had the chance to look at rock specimens and minerals under microscopes, in order to see crystal growth and other fine details.
4. The final activity involved the students repeatedly rolling chunks of playdoh, beads, and leaves down the previously created cardboard mountains, into a large bin, followed by pieces of cardboard placed on top to form layers. They then took turns stepping down on the layers with their feet, demonstrating the processes of erosion and compaction.
5. A questionnaire related to rocks was given to students after the lecture and hands-on activities. Students were asked to identify which rocks were sedimentary using everything they were exposed to in the lesson. The students were not told which rocks they had seen and not seen at this point of the lesson, and were intended to rely solely on the lecture and activities they had done in class with us. The examples in these questions will be analyzed as "active" and "inactive" examples respectively. As a reward for their participation, each student received an amethyst keychain as a souvenir of the lesson.

Background:

The purpose of this study was to teach kindergarten and 2nd-grade students about Earth's natural processes through hands-on and engaging activities. We aimed to gauge the effectiveness of a more interactive and visual teaching style in the classroom, and compare the results with standard teaching methods that are not as engaging. The experiment took place over the course of a school year (repeated twice), in an elementary school. We had the pleasure of teaching classes of K-2 children about rocks and other earthly processes, while hopefully fostering a lasting interest in earth science for these students.

Anonymous Questionnaire:

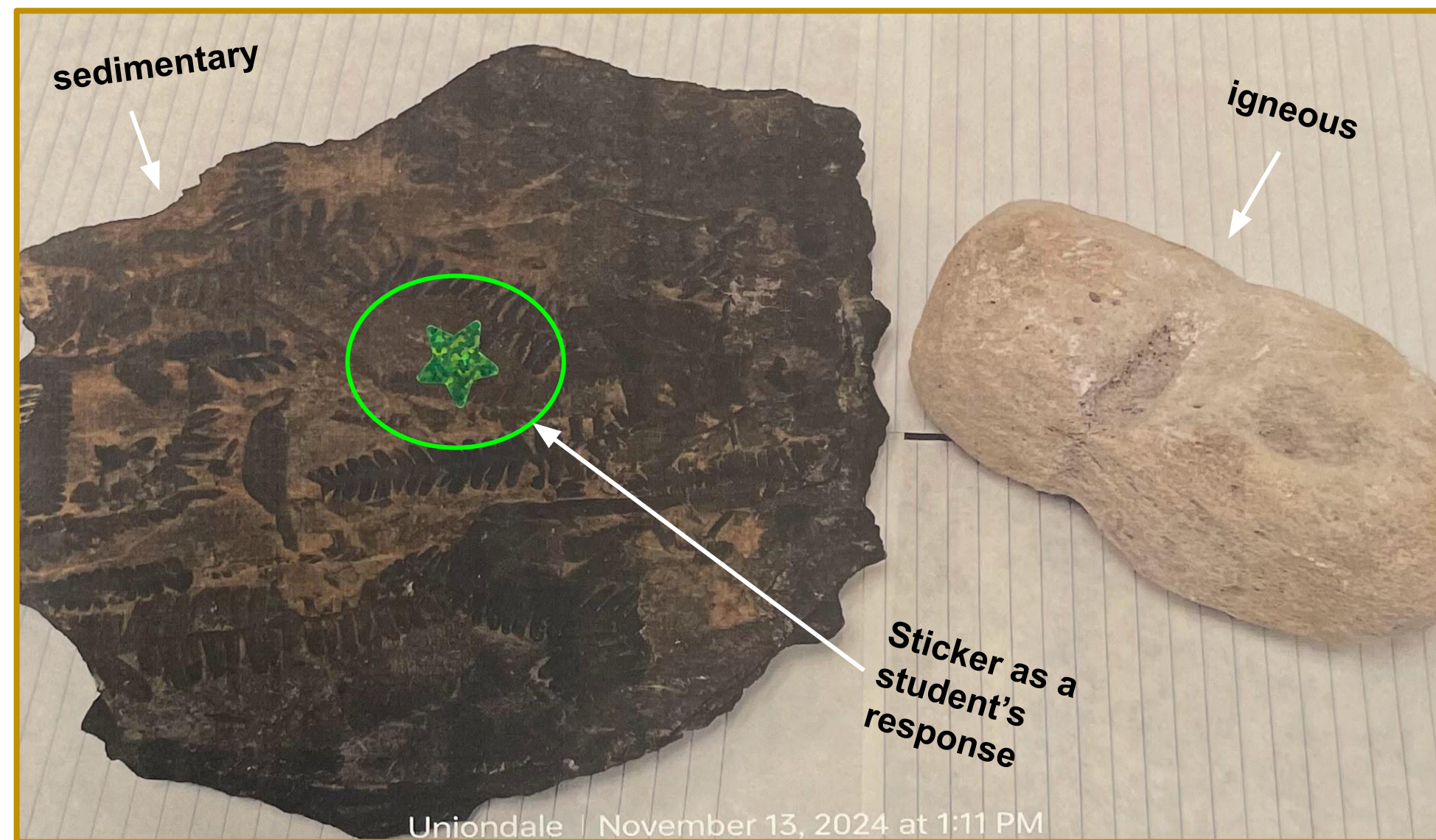


Figure 1: Photo of an answered question to an anonymous exam given to the students following the lesson, the green sticker reflects the rock that the student believed was the sedimentary rock. This question was based on **active** teachings.

Figure 2: Photo example of the mini conglomerate rocks made by 2nd graders for one of our activities.



Figure 3: Photo of kindergartners looking at rock and mineral samples underneath a microscope.



Figure 4: Photo of the kindergarten class taken after the cardboard and playdoh activities, as they answer the anonymous questionnaire seen in Figure 1.

Results:

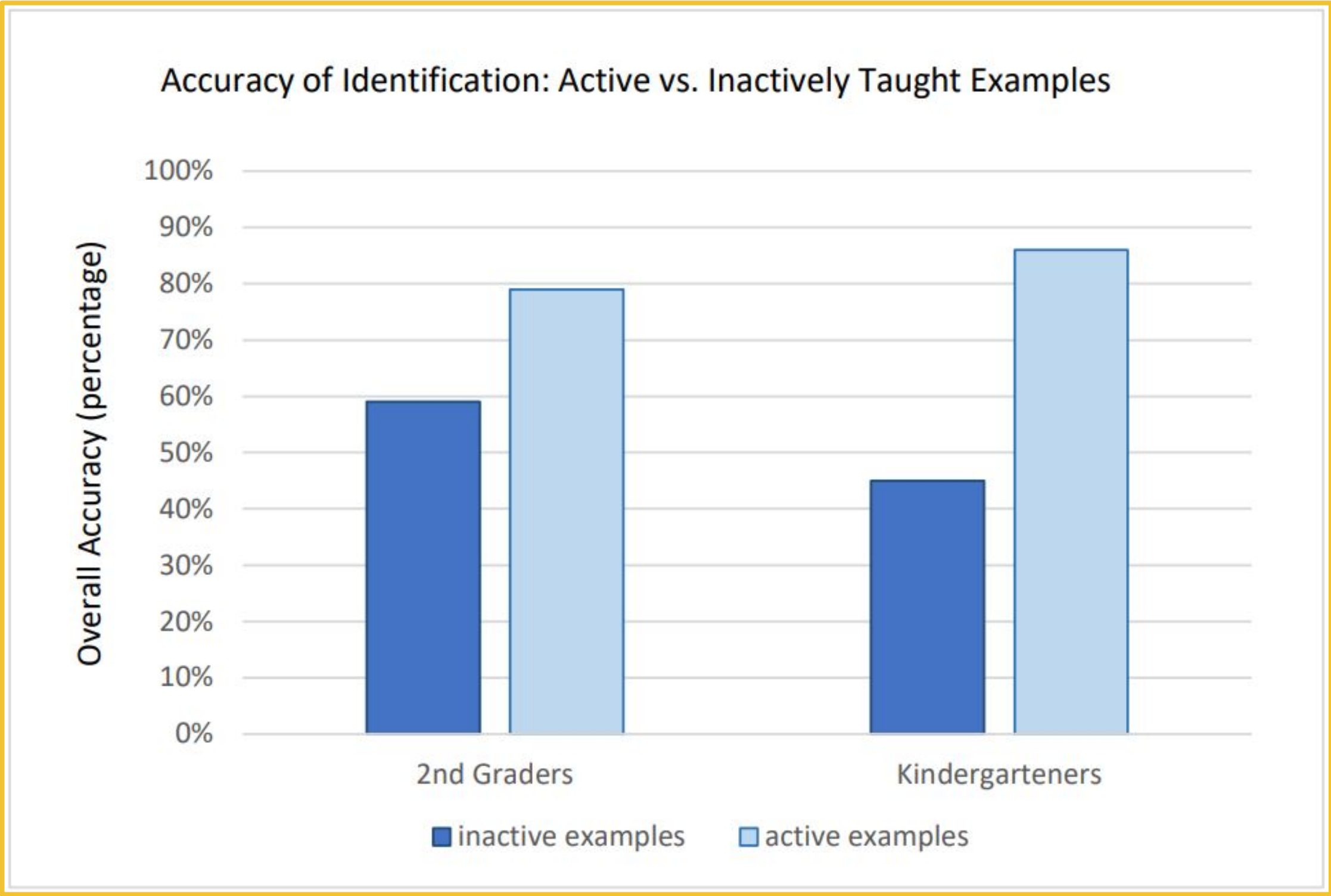


Figure 5: Bar graph comparing the students' sedimentary rock identification accuracies of rocks they were **actively** exposed to, and rocks that they were **inactively** exposed to.

After analyzing the responses of the anonymous questionnaires, the following observations can be made. For the class of 27 second graders, it can be seen that they had an overall accuracy of 59% in correctly identifying the sedimentary rocks that they were **inactively** exposed to. On the contrary, they had an overall accuracy of 79% in correctly identifying the sedimentary rocks that they were **actively** exposed to. There is a 20% difference between their inactive and active example accuracies.

For the class of 47 kindergartners, it can be seen that they had an overall accuracy of 45% in correctly identifying the sedimentary rocks that they were **inactively** exposed to. However, they had an overall accuracy of 86% in correctly identifying the sedimentary rocks that they were **actively** exposed to. Their results show a 41% difference in accuracy between active and inactive examples. This difference is just over twice as much as the 2nd graders.

AI Generated Image Example:

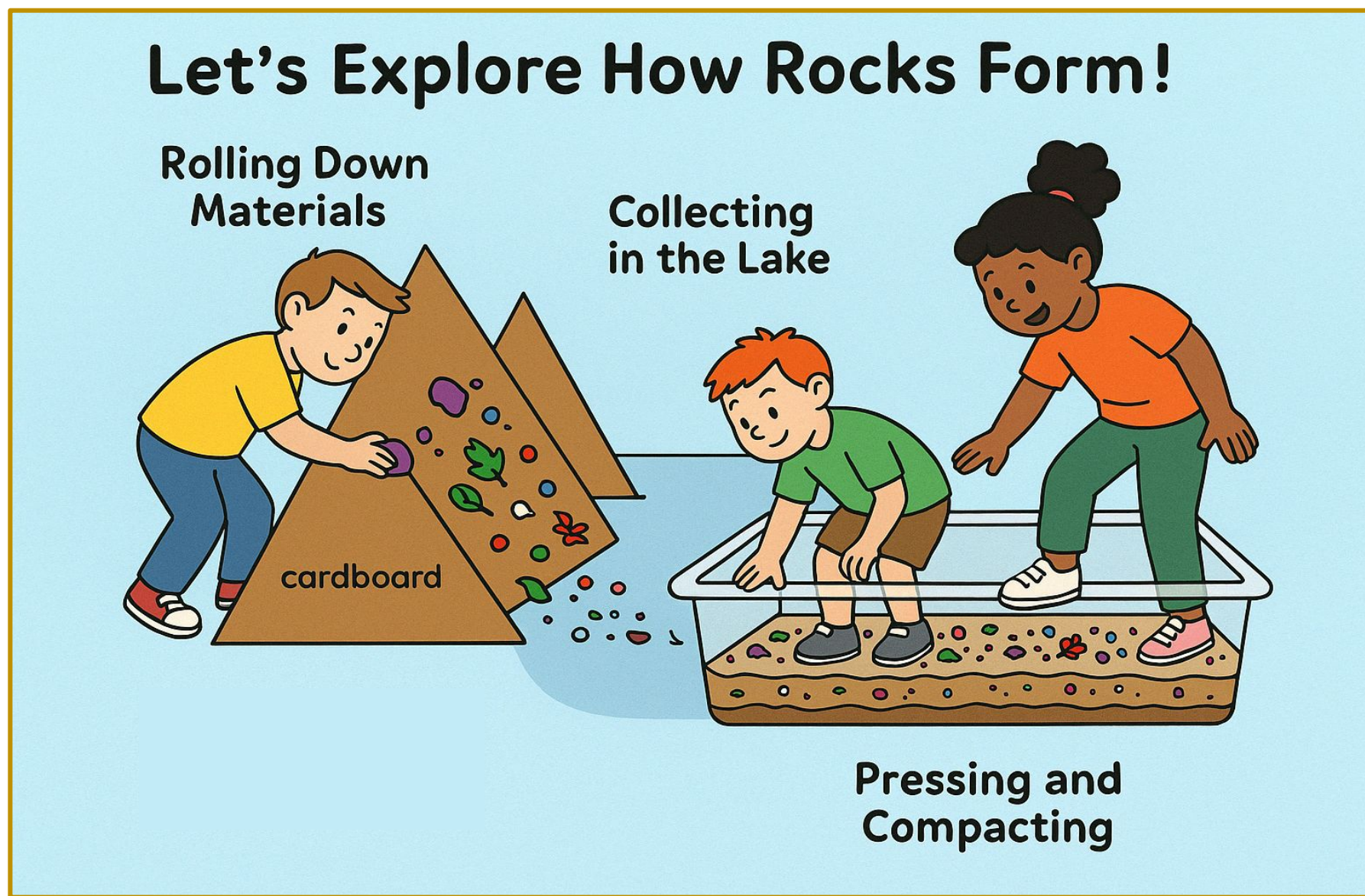


Figure 6: An AI-generated image based on a previously generated description of one of the activities. It effectively visualized the process, and was shown to students in advance.

Discussion + Conclusion:

The results show a clear indication that active teaching methods are more effective in educating students than inactive teaching methods. As seen in figure 5, there was a higher accuracy rate in identifying the sedimentary rocks that were **actively** taught, among both the second graders and the kindergartners. Combining this with the previously shown AI-generated images (Figure 6), other child-friendly visualizations, along with repetition of the lesson, led to a higher retention rates of the knowledge extended to the students.

When reviewing the differences in accuracy between active and inactive examples, The second graders had a smaller difference between their accuracies than the kindergartners. This could be due to the fact that they are older, with an age range of 6-7, and have slightly higher levels of critical thinking that they applied to their identifications. The larger difference in accuracies in the kindergartener's results could be attributed to their younger age, with their ages ranging from 4-5 years old. This causes kindergartners to be more affected by different teaching methods. They were highly receptive of the **actively** taught information, and not very receptive of the **inactively** taught information. This observation emphasizes the need for more active approaches in the kindergarten grade especially.

There were limitations in the collecting of the data used for this project. Around 20 questionnaires were omitted from the data that was used to create Figure 5, due to incomplete or grouped responses that would have skewed the data. Other limitations include lack of time; we were only allotted one hour per grade to teach everything. Additionally, not all of our AI generated images could be used due to scientific inaccuracies that may stem from AI's limited image creation capabilities at this time. Despite this, our findings are not disproven and still reflect the valid data collected.

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